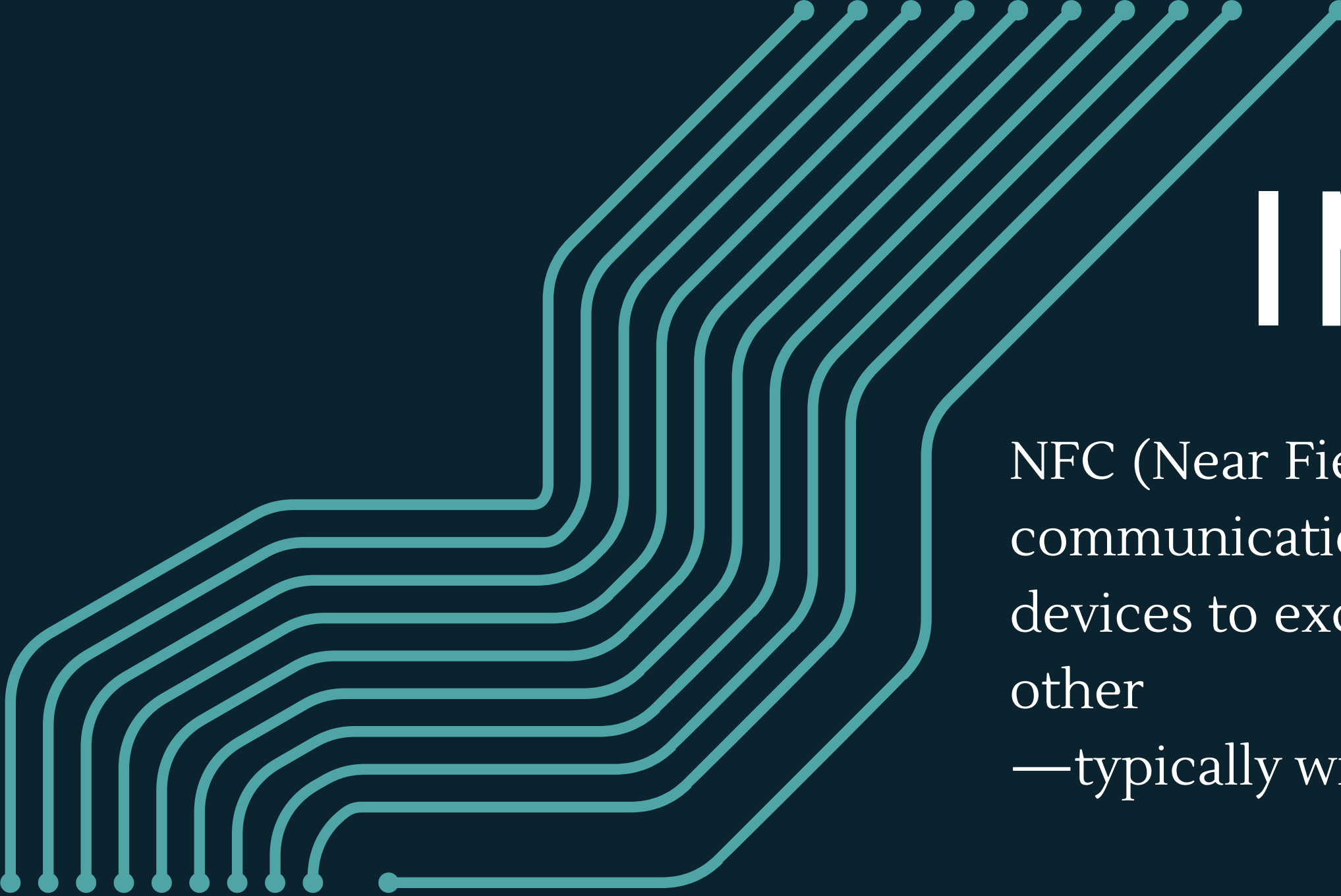


NFC (NEAR FIELD COMMUNICATION)





INTRODUCTION

NFC (Near Field Communication) is a short-range wireless communication technology that allows two electronic devices to exchange data when they are very close to each other
—typically within 4 cm or less.

What it means

- Near Field → very short distance communication
- Communication → exchange of data between devices

So, NFC = communication between devices at a very close distance.



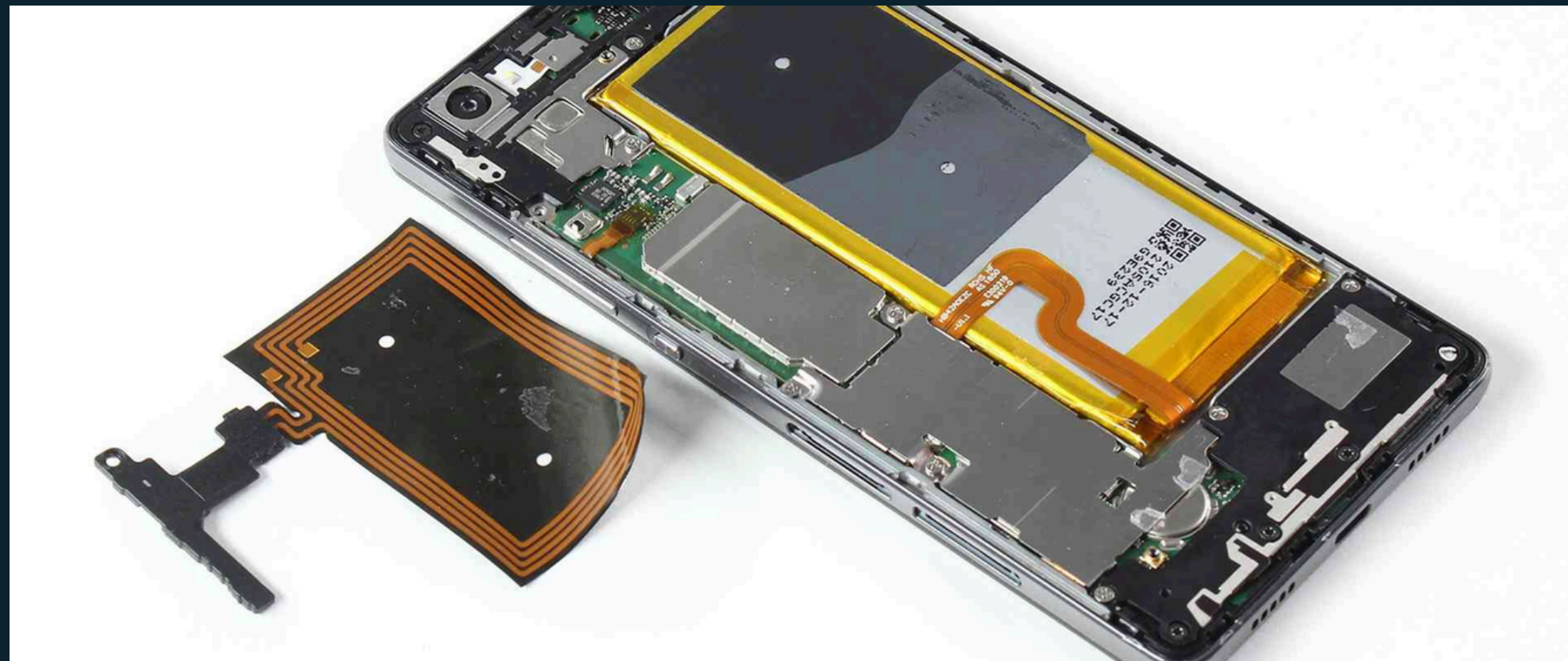
HOW IT WORKS



NFC is based on electromagnetic induction. When two NFC-enabled devices come close:

- An NFC device has a coil antenna
- When current flows through this coil, it creates a magnetic field
- A nearby coil (in another device) picks up this field and converts it back into electrical signals

COIL ANTENNA

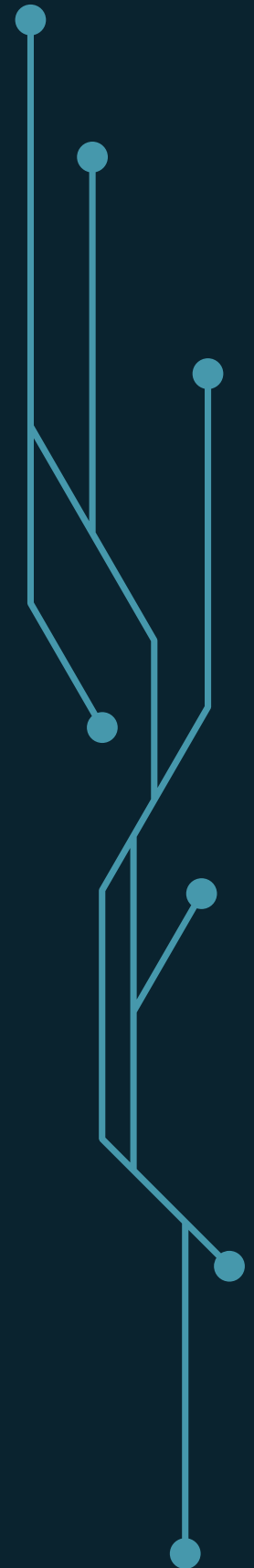


TWO MODES OF OPERATION

(A) Passive Mode

- Only one device is powered (like your phone)
- The other device (like a card or tag) has no battery
- The powered device creates the magnetic field
- The passive device uses that energy to send back data

Example: tapping a metro card



TWO MODES OF OPERATION

(B) Active Mode

- Both devices generate their own fields
- They take turns sending and receiving data

Example: phone-to-phone data transfer





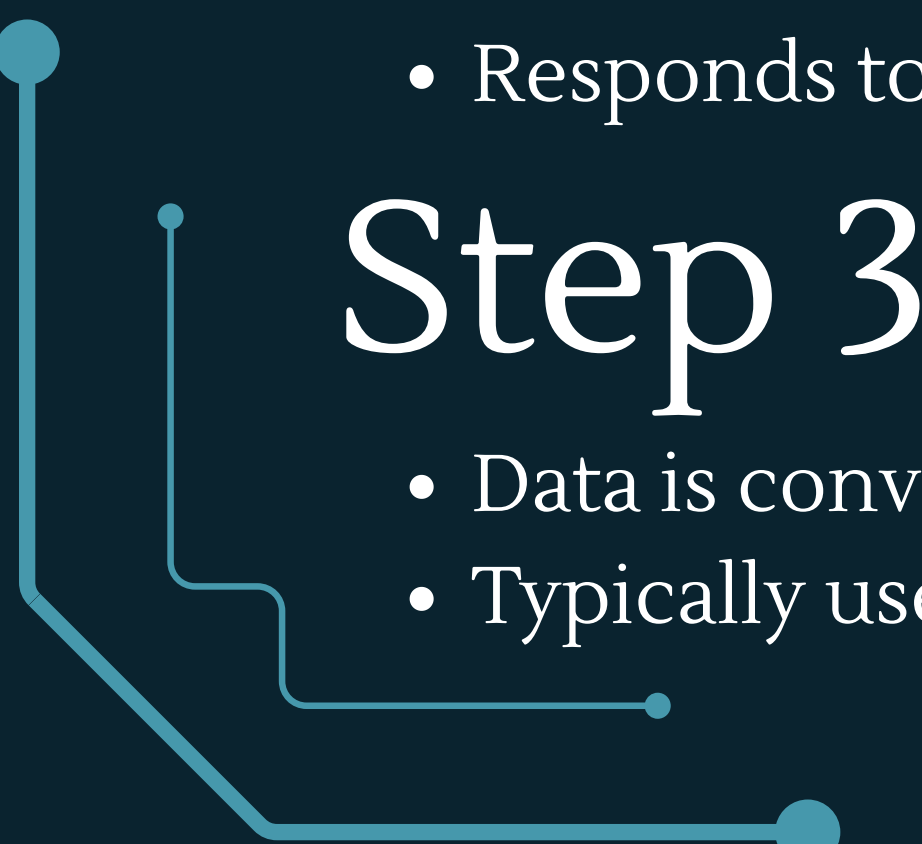
Step 1: Field Generation

- Initiator device (e.g., phone) generates a 13.56 MHz RF field
(This is the standard NFC frequency)

Step 2: Device Detection

- When another NFC device enters the field:
- It detects the signal
- Responds to the initiator

Step 3: Data Encoding

- Data is converted into signals using modulation techniques
 - Typically uses Amplitude Shift Keying (ASK)
- 



Step 4: Data Transfer

- Data is sent in small packets
- Speed ranges:
 - 106 kbps
 - 212 kbps
 - 424 kbps



Step 5: Response (Load Modulation)

- Passive device sends data back using load modulation
 - It slightly changes the load on its antenna
 - This change is detected by the initiator
- This is a clever trick: no need for a battery!

Common uses of NFC

- Contactless payments (like Google Pay, Apple Pay)
- Data sharing between smartphones
- Access cards (ID cards, metro cards)
- Smart tags (tap to perform actions like turning on Wi-Fi)



THANK YOU

